

BIOLOGY (044)

Reconstructed Question Paper — 2025

Reconstructed from the submitted answer booklet. Where the original printed wording is not visible, the question has been reconstructed from the corresponding answer and topic rather than invented as an exact quotation.

Section A — Multiple Choice Questions

Q.1–Q.16 Answer the following multiple-choice/assertion–reason questions based on the prescribed Biology syllabus. The answer booklet shows questions involving:

1. 23S rRNA and ribosomes,
2. genetics and inheritance,
3. X-linked recessive traits,
4. human reproduction,
5. biotechnology and molecular biology,
6. ecological relationships,
7. reproductive biology and conservation.

Section B — Short Answer Questions

Q.17 Explain the structure of the double-stranded DNA molecule. Include the number of base pairs per turn, the distance between successive base pairs and the role of hydrogen bonding.

Q.18 Explain the legal/medical provisions related to termination of pregnancy in India. State the major safeguards and conditions under which medical termination of pregnancy is permitted.

Q.19 Explain decomposition. Describe its major steps:

- (i) fragmentation,
- (ii) leaching,
- (iii) catabolism,
- (iv) humification,
- (v) mineralisation.

Explain the significance of humus.

Q.20 Explain the use of restriction enzymes in recombinant DNA technology. Why should the same restriction enzyme be used to cut the vector and foreign DNA? Explain the role of DNA ligase.

Q.21 Explain the importance of colostrum and why it is considered important for the newborn. Mention the major antibodies present and the protection they provide.

Section C — Long Answer Questions

Q.22 Explain the biological treatment of sewage using activated sludge. Describe:

- (a) the formation of flocs,
- (a) the role of aerobic microbes,
- (a) return of a portion of activated sludge,
- (a) digestion in anaerobic sludge digesters,
- (a) production/use of biogas.

Q.23 What is self-incompatibility in flowering plants? Explain how it prevents self-pollination/self-fertilisation and promotes outbreeding. State its evolutionary significance.

Q.24 Explain the impact of alien species invasions. Discuss examples such as Nile perch and Parthenium and their effects on native biodiversity.

Q.25 Explain the concept shown in the genetic/reproductive problem involving birds and determine the sex-chromosome constitution of male and female birds.

The remaining questions, diagrams and exact MCQ choices are not fully recoverable from the answer booklet.